



TU Delft Aerospace Engineering

Electricity and Magnetic Fields

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The main concepts

- Current [Amps, A]
 - Charge [Coulombs, C]
 - Energy [Joules, J]
 - Power [Watts, W]
 - Voltage [Volts, V]
 - Resistance [Ohm, Ω]
-
- Time [seconds, s]

Current, I [A]

Flow of electric charge

Remember mass flow? Well current is the derivative of charge (with respect to time)
(Imagine charge as water, then current is the charge flow)

So $I = Q/t$

Charge, Q [C]

How much electric 'stuff' something has

Its like mass but for electricity

If a fluid travelling through a hose is very dense, the mass is large → Mass flow is large

Same for electricity

If the charge is dense, the “electric mass” is very large → Electric flow (Current) is large

The charge of one electron is defined as $e = 1.6 \cdot 10^{-19} \text{ C}$

Energy, E [J]

How much work electric current can do

How much light it can deliver to a bulb, or how much it can heat it up

It usually comes from:

- Batteries (Chemical Energy)
- Power plants (Thermal/ Mechanical Energy)
- Solar panels (light Energy)

Power, P [W]

Energy flow

How much energy flows through a point per second

Usually we measure our devices with this - how much electric energy flows from the socket to my phone per second - the larger the better

How fast electric energy is transferring energy

The efficiency η_{electric} is the ratio of useful power output to total power input: $P_{\text{out}}/P_{\text{in}}$

Voltage, V [V]

Potential difference between two points

Think of it as “push” - how much the charges are being pushed

Similar to pressure in a hose. If the pressure difference / potential difference is very large, the push will be very large

It is sometimes called electric potential - the energy per unit charge:
How much energy each unit of charge has - $V = E / Q$

Resistance, R [Ω]

How much a material resists electricity

Like friction!

The larger the resistance, the smaller the flow of electricity

Or how narrow a pipe of water is - the narrower, the more difficult it is for electricity to flow through

Ohm's Law: $V = I R$ - relationship between voltage, current and resistance

Relationships (Formulae)

$$\text{Energy [J]} = \text{Power [W]} * \text{time [s]} \quad [\text{J}] = [\text{W s}]$$

$$\text{Electrical Energy [J]} = \text{Electric Charge [C]} * \text{Voltage [V]} \quad [\text{J}] = [\text{C V}]$$

$$\text{Electric charge} = \text{Current [A]} * \text{time [s]} \quad [\text{C}] = [\text{A s}], [\text{J} = \text{A V s}]$$

$$\text{Electrical Energy [J]} = \text{Current [A]} * \text{time [s]} * \text{Voltage [V]}$$

$$\text{Voltage [V]} = \text{Current [A]} * \text{Resistance } [\Omega]$$

Questions

- 1) A smartphone charger delivers a steady current of 2.0 A to a battery. If the phone is left to charge for 30 minutes, calculate the total electric charge that has passed into the battery.
- 2) An electric heater is connected to a 230 V supply and has an internal resistance of 46 Ohms .
 - a) Calculate the current (I) flowing through the heater.
 - b) Determine the power (P) of the heater in Watts (W).
- 3) A lightning bolt transfers 15 C of charge through a potential difference of $100,000,000\text{ V}$. How much electrical energy (J) is released during this strike?

Questions

4) A high-performance electric motor is designed such that its power consumption $P(t)$ increases linearly with time to overcome initial inertia, following the function $P(t) = 50t$.

- a) What is the total Electrical Energy consumed between $t = 0$ and $t = 10$ seconds.
- b) If the motor operates at a constant voltage of 25 V, determine the instantaneous Current flowing through the motor at exactly $t = 10$ seconds.

Electric circuit

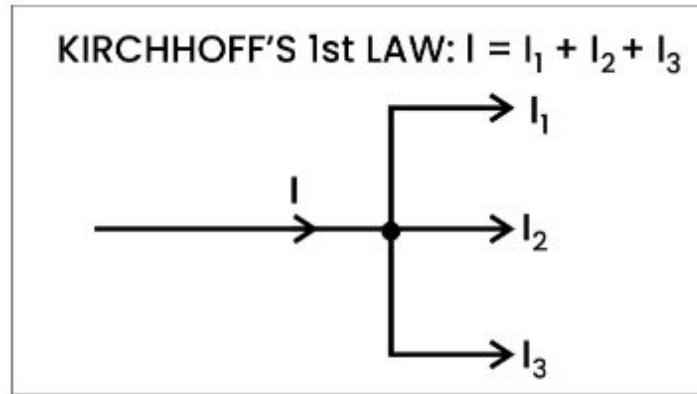
Water Analogy Refresher:

- Voltage = Water pressure
- Current = Flow rate of water
- Resistance = Narrowness/friction in the pipe
- Junction/Node = Where pipes split or join
- Loop = A closed path for water to flow

Kirchhoff's Current Law– “What goes in must come out”

The flow going in is the same as the flow going out

If a wire splits into two, they will share the flow (current)



Kirchhoff's Voltage Law– “Energy is conserved around a loop”

Imagine in a water circuit, the components need to drink water to work.
So when the water gets to them, they take some.

After the component, the amount of water reduces, so does the pressure and the energy

And so on.

After every component, the energy drops, until it reaches 0 in the final component.

Kirchhoff's Voltage Law– “Energy is conserved around a loop”

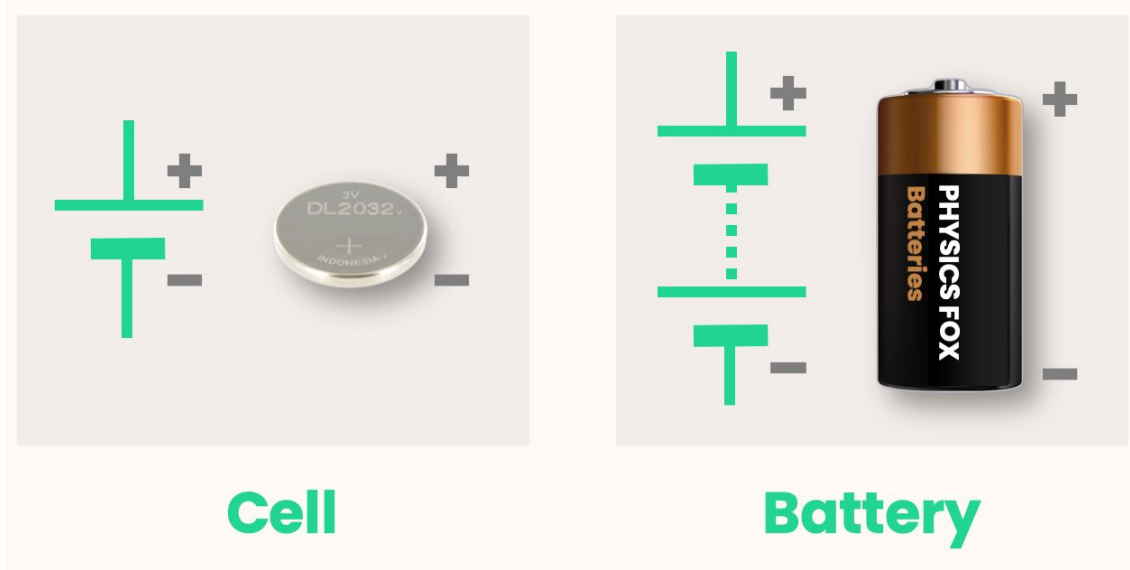
For electricity it is similar

After every component, the potential difference drops

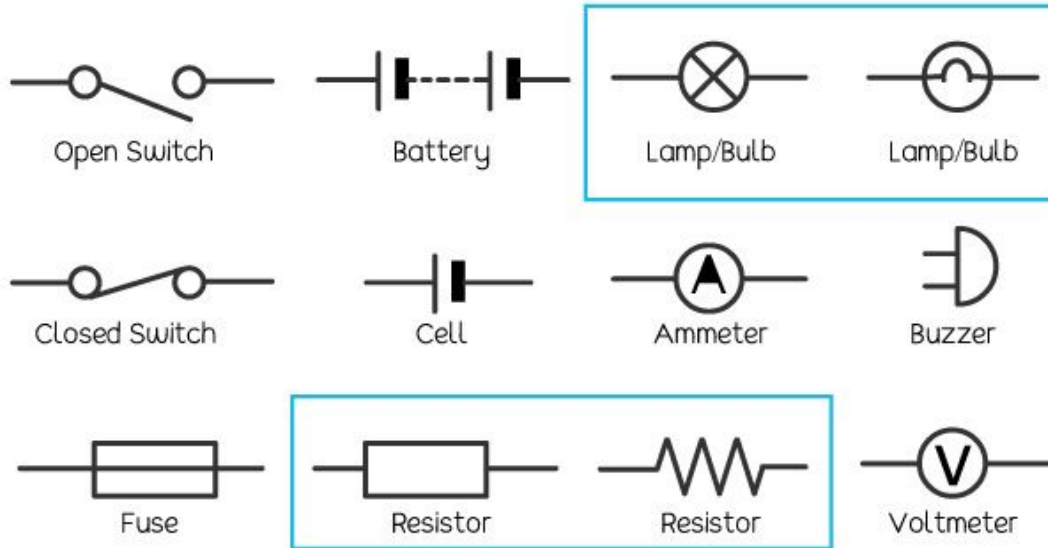
It keeps dropping after each component until at the very end the potential difference is 0

Then it is “recovered at the cell”

Cell and battery



Other components



Voltage in parallel

In parallel, the components are connected to the same 2 points

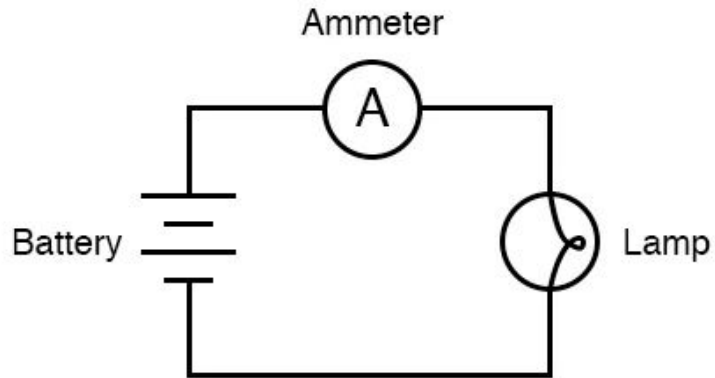
So it makes sense that the voltage is the same for both

The potential difference between the two points they are connected to are the same

How to measure current

We want to see the current - flow

So we place it directly on the pipe / wire



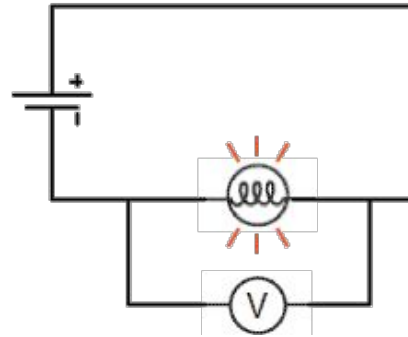
How to measure voltage

We want to see the voltage: we place it in parallel

A voltmeter has very high resistance, so most of the current goes through the other branch

Most of the charge flows through the other branch, so circuit is almost unaffected

So we get an accurate measure of the voltage



Why use parallel circuits?

Why use parallel circuits?

Same voltage across each component

- All devices (like light bulbs) will perform at their designed voltage.

Independent Components:

- If one component fails, it does not affect the others.

Better for Power Distribution

- Used in homes

Why use series circuits?

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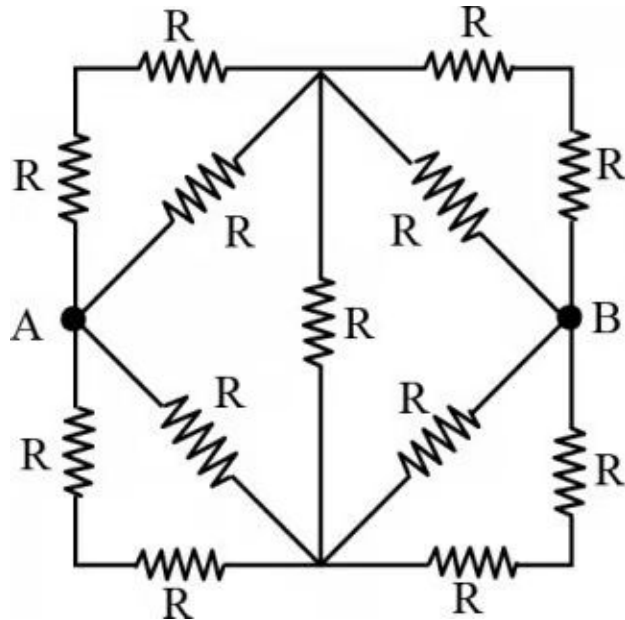
Simple and Easy to Build

Used for Specific Current Control

Used in flashlights, christmas lights, simple circuits

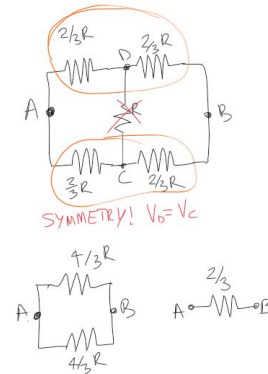
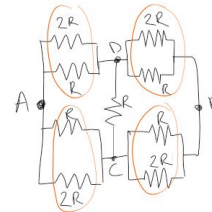
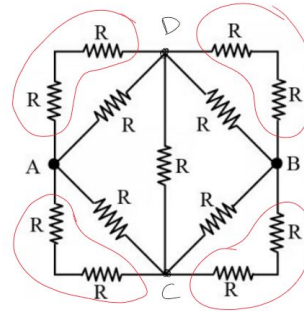
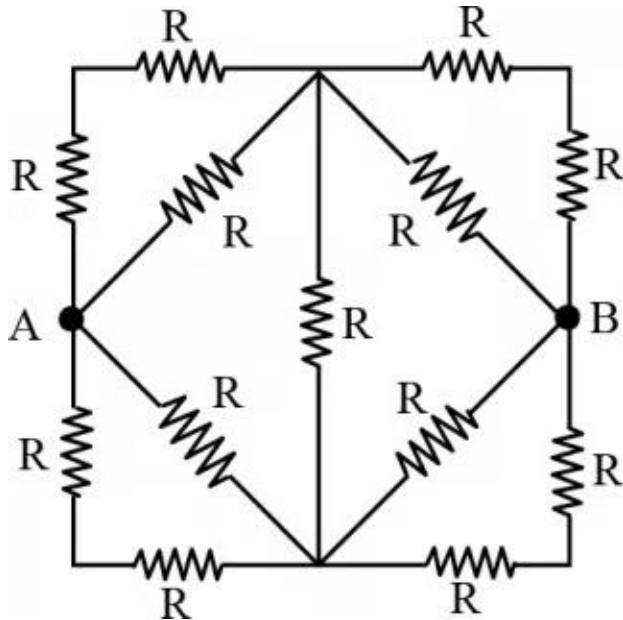
Exercises

Find the equivalent resistance between A and B



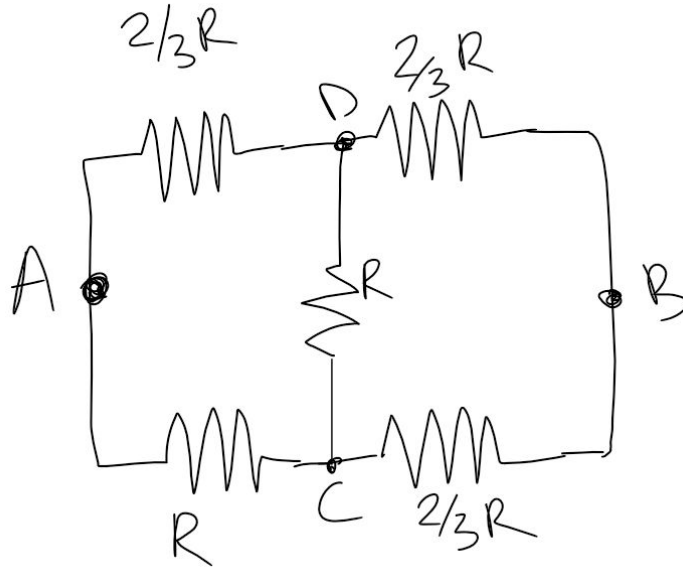
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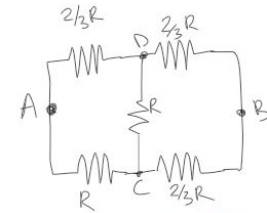
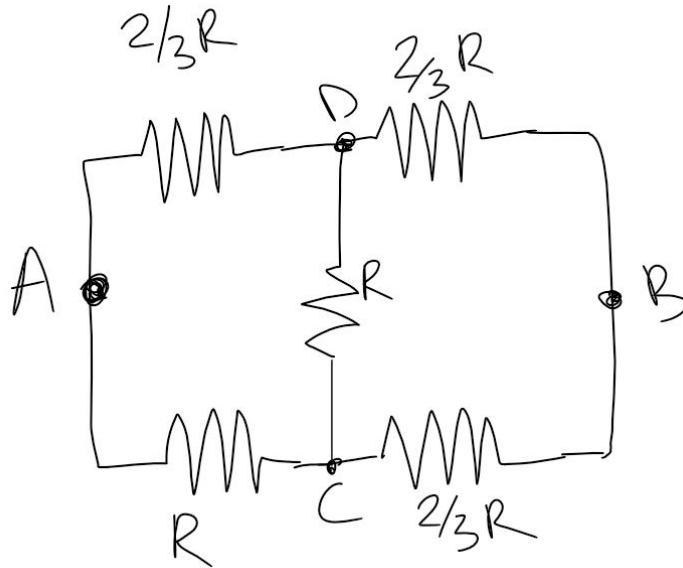
Exercises

Find the equivalent resistance between A and B



Exercises

Find the equivalent resistance between A and B



SET $V_A = 1V$, $V_B = 0V$

$$D) \frac{V_D - V_A}{\frac{2}{3}R} + \frac{V_D - V_B}{\frac{2}{3}R} + \frac{V_D - V_C}{R} = 0$$

$$C) \frac{V_C - V_A}{R} + \frac{V_C - V_B}{\frac{2}{3}R} + \frac{V_C - V_D}{R} = 0$$

$$V_D = \frac{25}{52}V, \quad V_C = \frac{11}{26}V$$

$$I_{tot} = \frac{V_A - V_D}{\frac{2}{3}R} + \frac{V_A - V_C}{R} = \frac{141}{104R}$$

$$R_{eq} = \frac{V_A}{I_{tot}} = \frac{104R}{141}$$

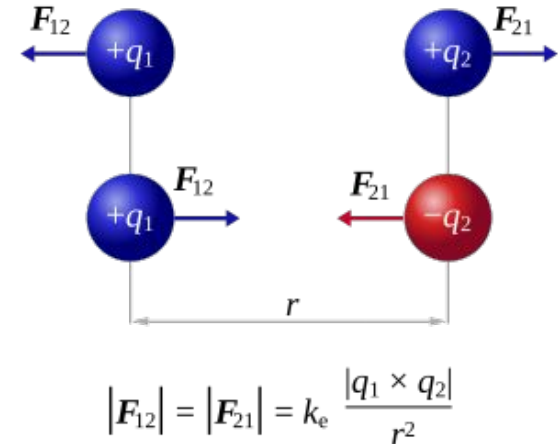
Coulomb's Law

Coulomb's Law is used to calculate the electrostatic force between two point charges separated by a certain distance - similar as with gravitational force

Opposites attract, likes repel

Same analogy as with the water - charge = mass

$$k_e = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$$



Coulomb's Law - Exercise

Two point charges, $q_1 = 2 \mu\text{C}$ and $q_2 = -4 \mu\text{C}$, are separated by a distance of 0.5 m. What is the magnitude of the electrostatic force between them?

If the distance between two identical point charges is tripled, by what factor does the electrostatic force between them change?

Two small spheres carry charges Q and $4Q$. Sphere A exerts a force F on sphere B. What is the magnitude of the force exerted by sphere B on sphere A?

A $3 \mu\text{C}$ charge is placed at the origin, and a $-3 \mu\text{C}$ charge is placed at $x = 2 \text{ m}$. Where could a third positive charge be placed so that the net force on it is zero?

Coulomb's Law - Exercise

Two point charges, $q_1 = 2 \mu\text{C}$ and $q_2 = -4 \mu\text{C}$, are separated by a distance of 0.5 m. What is the magnitude of the electrostatic force between them? **0.288 N**

If the distance between two identical point charges is tripled, by what factor does the electrostatic force between them change?

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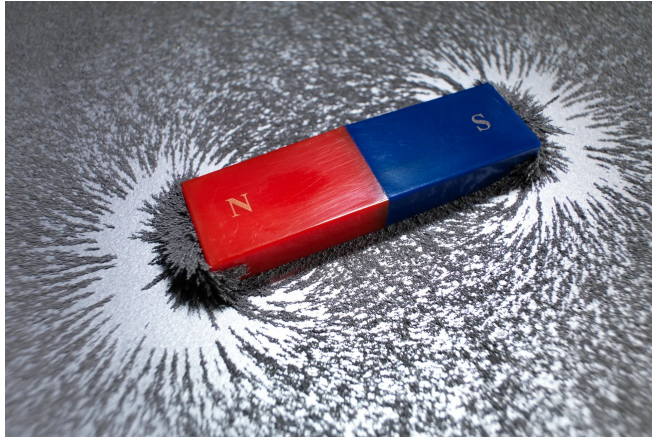
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A $3 \mu\text{C}$ charge is placed at the origin, and a $-3 \mu\text{C}$ charge is placed at $x = 2 \text{ m}$. Where could a third positive charge be placed so that the net force on it is zero? **There is no such point**

Magnetic Fields

Magnetic fields are regions where magnetic forces can be detected

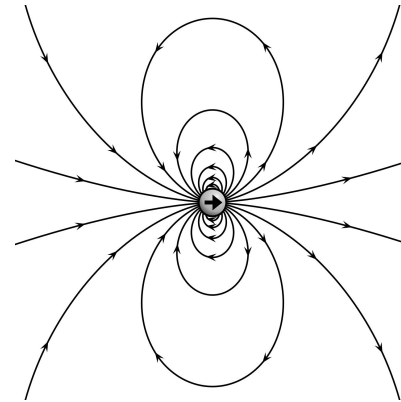
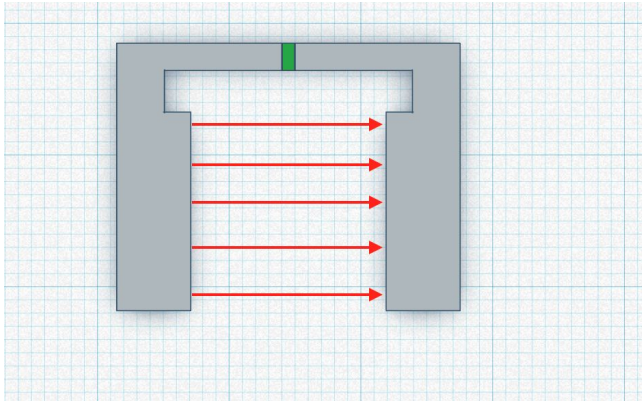
We plot magnetic field lines from North to South



Magnetic Fields

Magnetic fields are regions where magnetic forces can be detected

- **Homogeneous field:** The field strength and direction are constant at every point

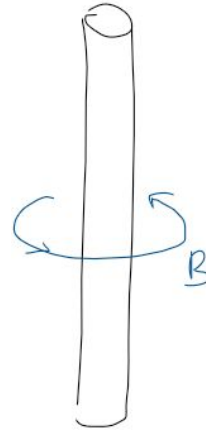


- **Inhomogeneous field:** Field strength and direction vary

Magnetic Fields

Magnetism is strongly linked to electricity. In fact, a wire carrying a current creates a magnetic field defined by the right hand rule.

So if the magnetic field of the following coil is anti clockwise, what is the direction of the current?



Magnetic Fields

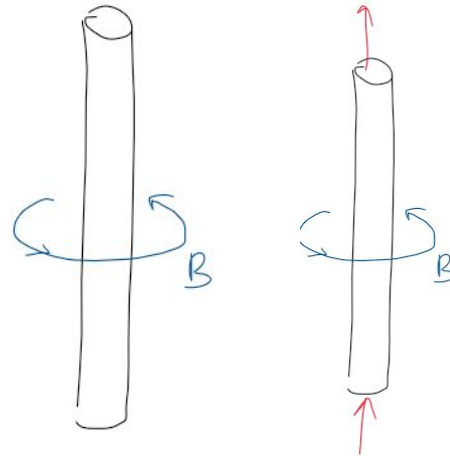
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So if the magnetic field of the following coil is anti clockwise, what is the direction of the current?

The strength of the magnetic field is given by $B [T] = (\mu_0 I) / (2 \pi R)$

where $\mu_0 = 4\pi \times 10^{-7} [T \cdot m/A]$
(permeability of free space)

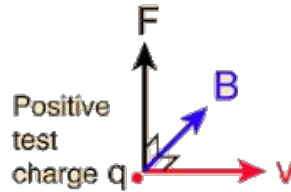
Notice how B changes with I and R !



Lorentz Force

A charged particle moving in a magnetic field experiences a force known as the Lorentz Force

$$\vec{F} = q\vec{v} \times \vec{B}$$



Determined by the right hand rule

Lorentz Force

1) A proton moves east with a speed of $4 \cdot 10^6$ m/s in a magnetic field of $2 \cdot 10^{-4}$ [T] directed into the page. What is the magnitude and direction of the magnetic force acting on the proton?

2) A 2.5 m long wire carries a current of 5.0 A in the presence of a magnetic field with a strength of $2 \cdot 10^{-3}$ [T].

Calculate the magnitude of the magnetic force on the wire.

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$1.28 \cdot 10^{-18}$ N upwards

2) A 2.5 m long wire carries a current of 5.0 A in the presence of a magnetic field with a strength of $2 \cdot 10^{-3}$ [T].

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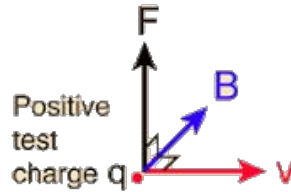
Calculate the magnitude of the magnetic force on the wire.

0.025 N

Lorentz Force

A charged particle moving in a magnetic field experiences a force known as the Lorentz Force

$$\vec{F} = q\vec{v} \times \vec{B}$$



Determined by the right hand rule

$$Q = I \cdot t, \text{ so } F = I \cdot L \times B$$

Lorentz Force

3) A 2.5 m long wire carries a current of 5.0 A in the presence of a magnetic field with a strength of $2 \cdot 10^{-3}$ [T]. The direction of the magnetic field is the same as the direction of the long wire.

Calculate the magnitude of the magnetic force on the wire.

Lorentz Force

3) A 2.5 m long wire carries a current of 5.0 A in the presence of a magnetic field with a strength of 2×10^{-3} [T]. The direction of the magnetic field is the same as the direction of the long wire.

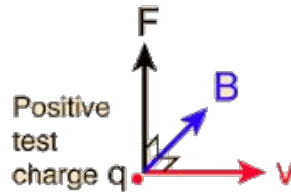
Calculate the magnitude of the magnetic force on the wire.

Magnetic field is parallel to the current, so the angle between the 2 vectors is 0° , hence the force is also 0

Lorentz Force

A charged particle moving in a magnetic field experiences a force known as the Lorentz Force

$$\vec{F} = q\vec{v} \times \vec{B}$$



Determined by the right hand rule

$$F = I \cdot L \times B = I \cdot L \cdot B \sin(\theta)$$

Magnetic Flux

Represents the total magnetic field passing through a given area. In a homogeneous field with magnetic strength B [T], then the magnetic flux is defined as:

$$\Phi_B = BA \cos(\theta)$$

Measured in Webers [Wb] = [T m²]

Magnetic Flux

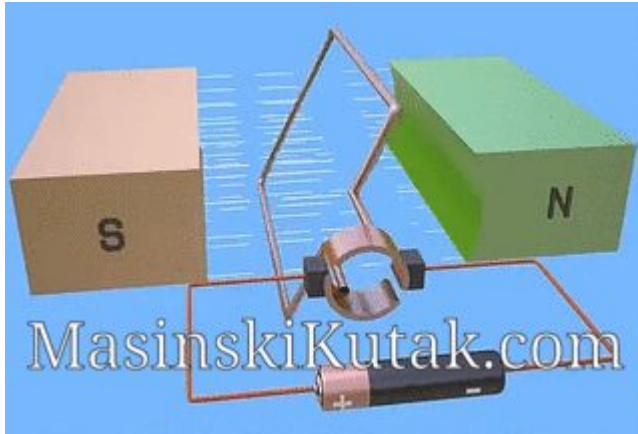
- 1) A magnetic field of 20 T is directed perpendicular to the face of a circular coil with a radius of 25 cm. Calculate the magnetic flux in this coil
- 2) What if the magnetic field were directed parallel to the face of the coil?
- 3) Imagine the coil were rotating, starting from being parallel to the magnetic field. Plot the variation of magnetic flux over a 180° revolution by finding the flux at 0, 30, 45, 60, 90, 135 and 180° .

Magnetic Flux

- 1) A magnetic field of 20 T is directed perpendicular to the face of a circular coil with a radius of 25 cm. Calculate the magnetic flux in this coil **3.93 Wb**
- 2) What if the magnetic field were directed parallel to the face of the coil?
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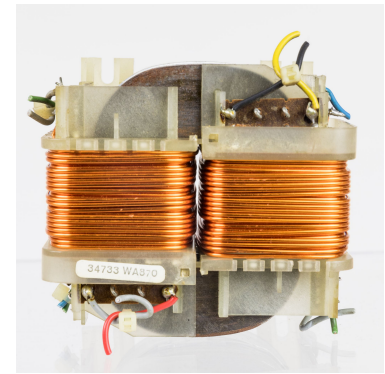
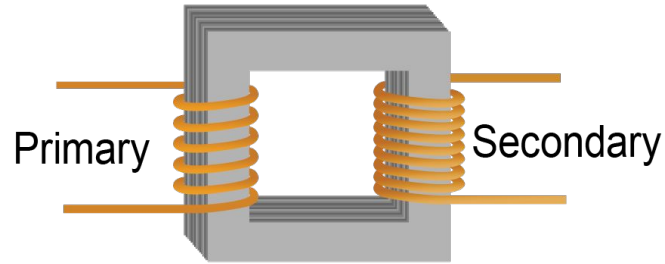
Induction

Faraday's Law of Induction: $\text{EMF} = -N * (d\Phi)/(dt)$



Induction

Transformers:



An alternating current flows through the primary coil. This generates an alternating magnetic field around it.

The iron core guides the changing magnetic flux into the secondary coil.

Because the secondary coil feels a changing magnetic field passing through its area, an electromotive force (Voltage) is induced across its ends.

Homework

Homework

1) An experimental conductor is cooled to cryogenic temperatures, causing its resistance to change as a function of the current passing through it, defined by $R(I) = R_0 + K I$, where $R_0 = 2 \text{ Ohms}$ and $K = 0.5 \text{ Ohms/Amps}$.

If a constant voltage of 12 V is applied across this conductor:

- a) Calculate the steady-state Current flowing through the circuit.
- b) Once the current is established, how many Electrons pass through a cross-section of the wire in 5 minutes?

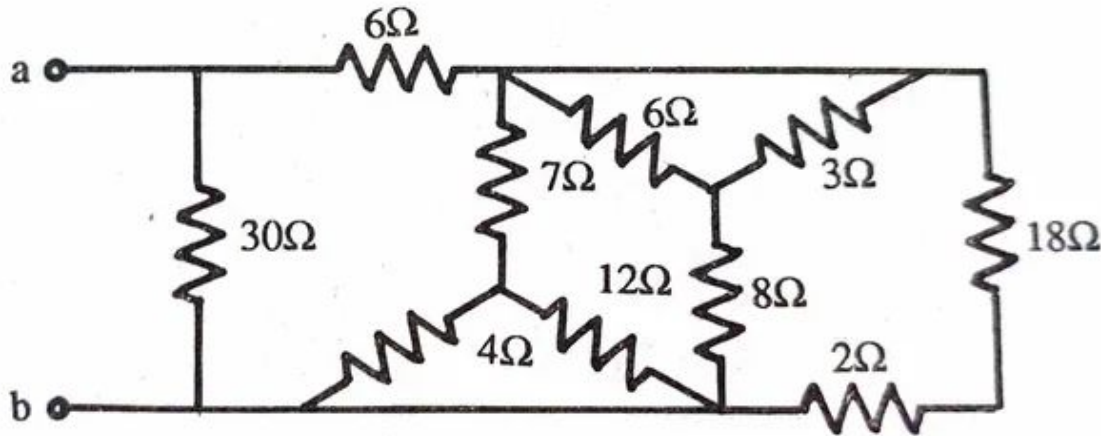
Homework

2) An electromagnetic railgun uses electrical energy to accelerate a 0.05 kg projectile. The system uses a massive capacitor bank that discharges a total Electric Charge of 200 C at an average potential difference of 5,000 V.

- a) Calculate the total Electrical Energy available in the discharge.
- b) If the system is only 15% efficient at converting electrical energy into the kinetic energy of the projectile, calculate the exit velocity of the projectile in meters per second.

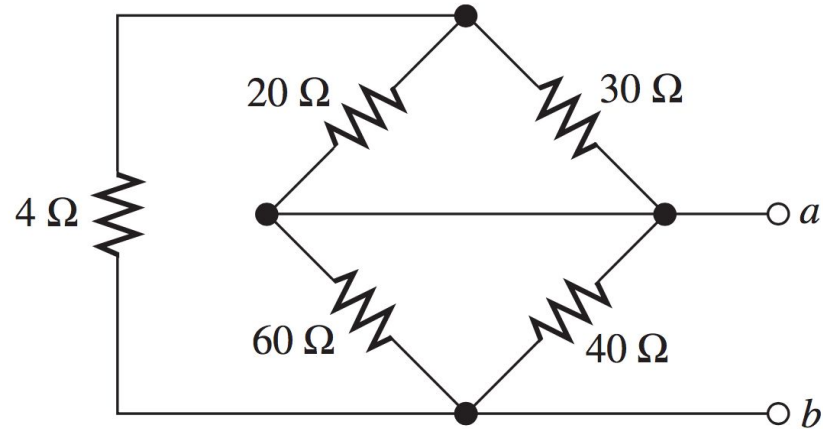
Homework

3) Determine the equivalent resistance between terminals a and b



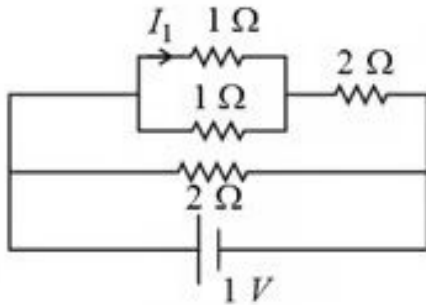
Homework

4) Find the equivalent resistance between terminals a and b



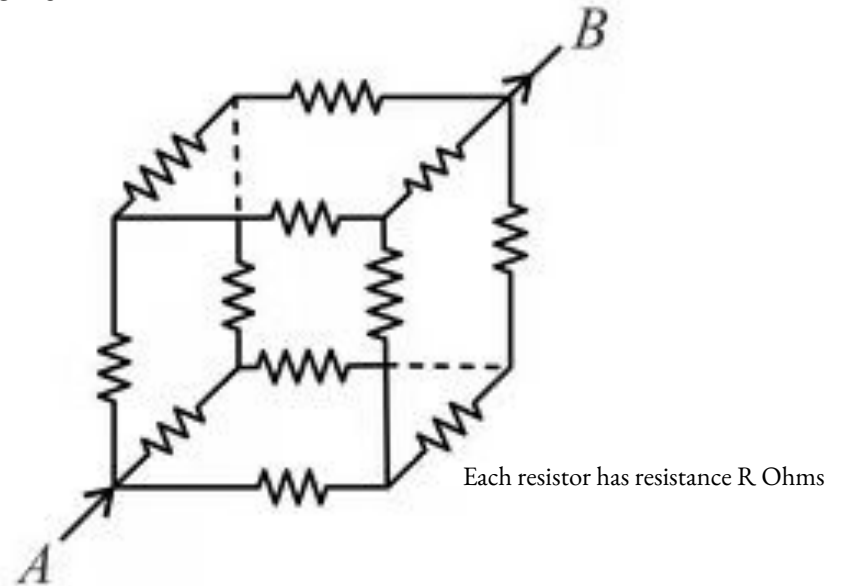
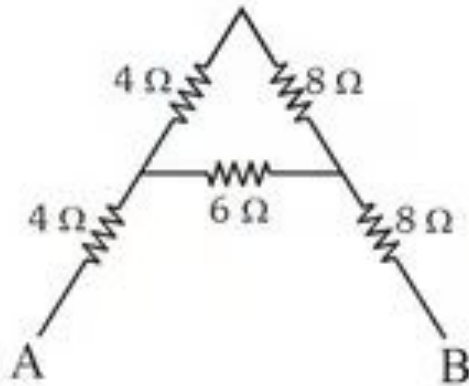
Homework

5) Find I_1 . What is the power dissipated by each of the 2 Ohm resistors?



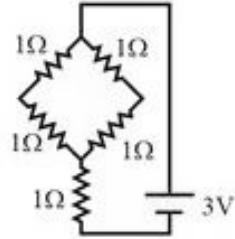
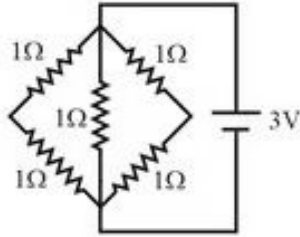
Homework

6) Find the equivalent resistance between A and B



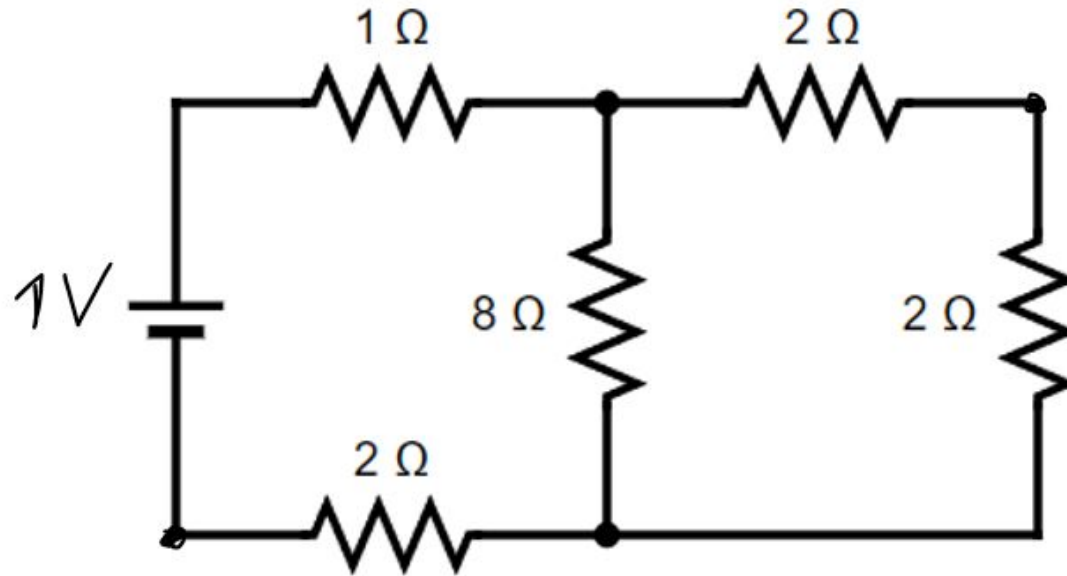
Homework

7) Which circuit dissipates the lowest power?



Homework

8) Find the voltage drop through each resistor using Kirchhoff's Law



Homework

9) Two identical small spheres, each with a mass of 2.0 g, are hung from a single point by two thin, non-conducting silk threads, each 1.0 m long.

The spheres are then given an equal positive electric charge Q .

Because they repel each other, the spheres come to rest in an equilibrium position where each thread makes an angle of 5° with the vertical.

- 1) Identify the three forces acting on one of the spheres. Draw a Free-Body Diagram
- 2) Calculate the distance between the centers of the two spheres when they are in this equilibrium position
- 3) Calculate the magnitude of the charge Q on each sphere.
- 4) If the entire setup were moved to the Moon (where gravity is approximately $1/6$ that of Earth) but the charge Q remained the same, would the angle increase or decrease? Explain your reasoning using the relationship between the forces.

Homework

- 10) A vertical wire carries a current of 45 A due south. Calculate the magnitude and the direction of the magnetic field 2 cm to the right of the wire.
- 11) A wire carries a current of 10 A. At what distance from the wire will a magnetic field of $8 \cdot 10^{-4}$ [T] be produced?

Homework

12)

René drops a magnet through a tube. Placed around the tube are three coils connected in series. See Figure 17. Using a computer, René measures the total voltage generated by the coils. Figure 18 shows the (induction voltage, time) graph that René got during this experiment. At $t=0.143$ s, the magnet is in the middle of the first coil. At that time, the induction voltage is zero.

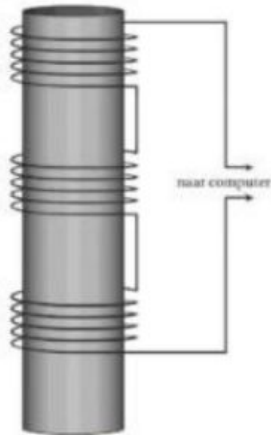


Figure 17

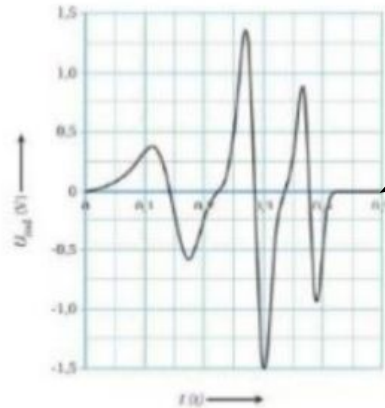


Figure 18

- a Explain that the induction voltage is zero when the magnet is passing the middle of a coil.

Homework

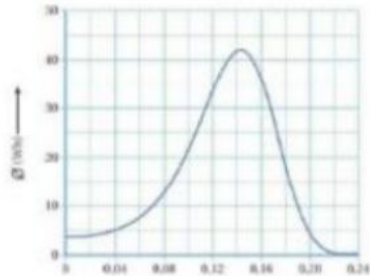
12)

At $t = 0.0$ s, René released the magnet. The magnet makes a free fall. The formula $v = g \cdot t$ applies for the falling magnet. The maximum induction voltage is directly proportional to the speed of the magnet.

The second coil has 600 windings.

b Determine the number of windings of the first and third coil.

The induction voltage at $t = 0.11$ s is smaller than the induction voltage at $t = 0.17$ s. The magnitude of the flux through the first coil as a function of time is shown in Figure 19.



c Explain that Figure 19 shows that the induction voltage at $t = 0.11$ s is smaller than the induction voltage at $t = 0.17$ s

Homework

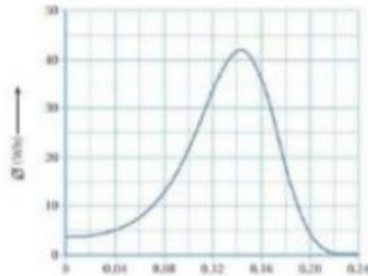
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c Explain that Figure 19 shows that the induction voltage at $t = 0.11$ s is smaller than the induction voltage at $t = 0.17$ s

Solutions

Homework

$$1) R(I) = R_0 + kI, \quad R_0 = 2, k = 0.5 \\ V = 12 \text{ V}$$

$$a) I = \frac{V}{R(I)} = \frac{12}{2 + \frac{1}{2}I}$$

$$I^2 + 4I - 24 = 0$$

$$I = \frac{-4 \pm \sqrt{16 + 96}}{2}$$

$$I = 3.29 \text{ A}$$

$$b) Q = It, \quad t = 300 \text{ s}$$

$$Q = 987.45 \text{ C}$$

$$e = 1.6 \cdot 10^{-19} \text{ C}$$

$$n = \frac{Q}{e} = 6.17 \cdot 10^{21}$$

$$2) m = 0.05 \text{ kg}$$

$$Q = 200 \text{ C}$$

$$V = 5000 \text{ V}$$

$$a) E_{\text{elec}} = QV = 1,000,000 \\ = 1 \text{ MJ}$$

$$b) \eta = \frac{E_k}{E_e} = 0.15$$

$$E_k = \eta E_e$$

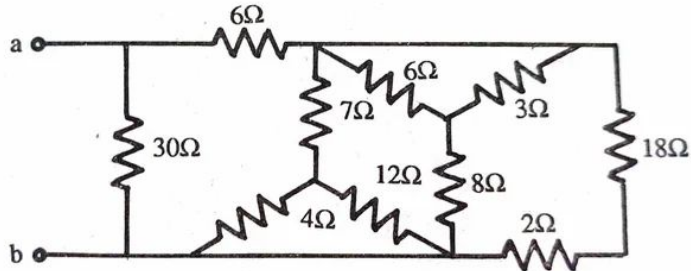
$$\frac{1}{2}mV^2 = \eta E_e$$

$$V = \pm \sqrt{\frac{2\eta E_e}{m}}$$

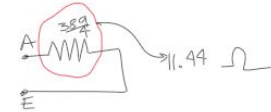
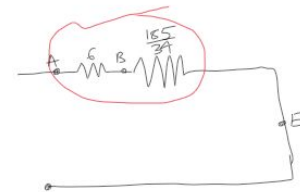
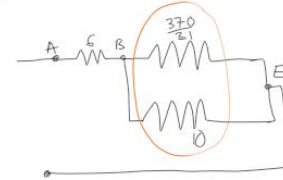
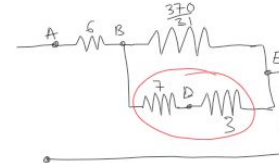
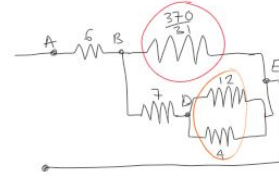
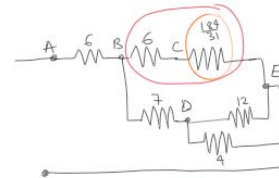
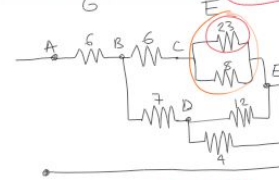
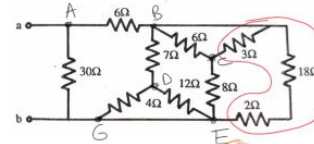
$$V = 2450 \text{ [m/s]}$$

Homework

3) Determine the equivalent resistance between terminals a and b

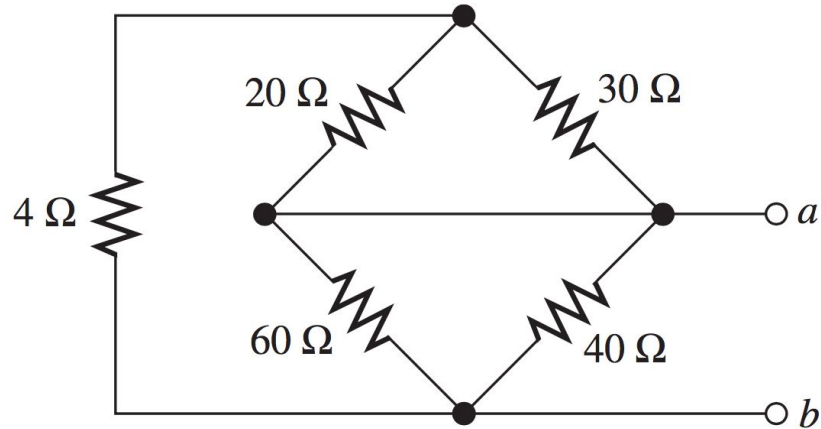


1) Determine the equivalent resistance between terminals a and b.

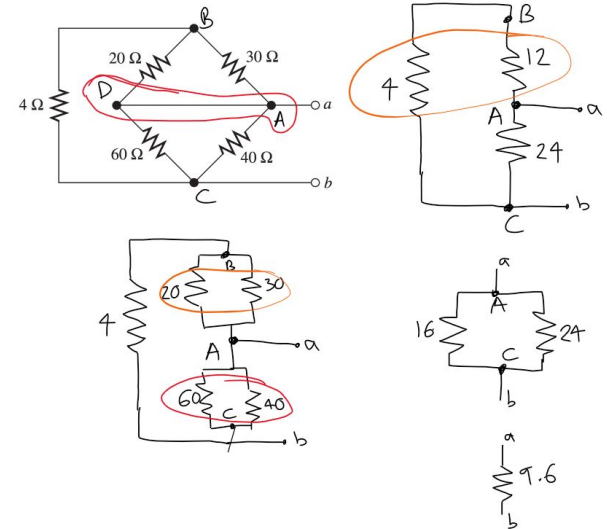


Homework

4) Find the equivalent resistance between terminals a and b

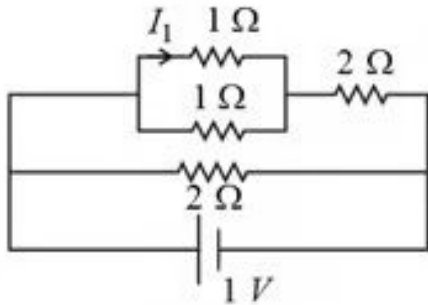


Find the equivalent resistance between terminals a and b in Figure P2.3.



Homework

5) Find I_1 . What is the power dissipated by each of the 2 Ohm resistors?



$$\Rightarrow) R_{eq} = \frac{1}{\frac{1}{1+1} + \frac{1}{2}} = \frac{10}{9}$$

$$I_{tot} = \frac{V}{R_{eq}} = \frac{9}{10} \text{ A}$$

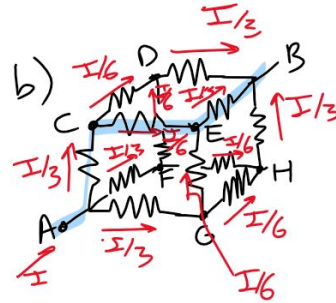
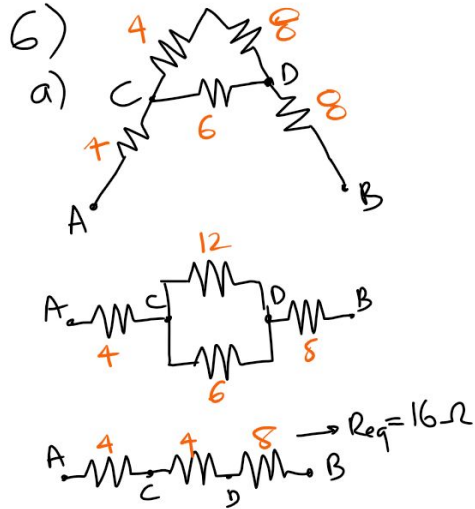
$$I_2 = \frac{1}{2}$$

$$I_1 = \frac{I_{tot} - I_2}{2} = 0.2 \text{ A}$$

TOP RESISTOR:
 $P = IV = I^2 R$
 $I = 2I_1 = 0.4$
 $P = 0.32 \text{ W}$

BOTTOM RESISTOR:
 $P = I_2 V = 0.5 \text{ W}$

Homework



FOLLOWING BLUE PATH:

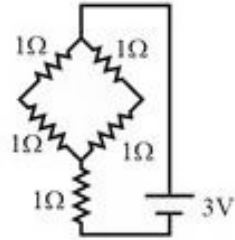
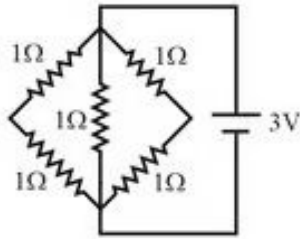
$$\Delta V = \frac{I}{3}R + \frac{I}{6}R + \frac{I}{3}R$$

$$\Delta V = \frac{5}{6}IR$$

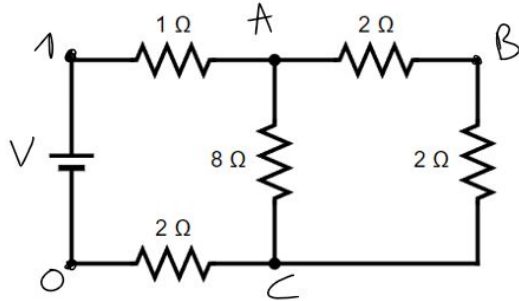
$$R_{eq} = \frac{\Delta V}{I} = \frac{\frac{5}{6}IR}{I} = \frac{5}{6}R$$

Homework

7) Which circuit dissipates the lowest power?



Homework



$$A) \frac{V_A - 1}{1} + \frac{V_A - V_B}{2} + \frac{V_A - V_C}{8} = 0$$

$$B) \frac{V_B - V_A}{2} + \frac{V_B - V_C}{2} = 0$$

$$C) \frac{V_C - V_B}{2} + \frac{V_C - V_A}{8} + \frac{V_C}{2} = 0$$

$$\textcircled{1} 13V_A - 4V_B - V_C - 8 = 0$$

$$\textcircled{2} -V_A + 2V_B - V_C = 0$$

$$\textcircled{3} -V_A - 4V_B + 9V_C = 0$$

$$\textcircled{2} -V_A = -2V_B + V_C$$

$$\textcircled{3} -6V_B + 10V_C = 0$$

$$V_B = \frac{5}{3}V_C$$

$$\textcircled{1} 26V_B - 13V_C - 4V_B - V_C - 8 = 0$$

$$22V_B - 14V_C - 8 = 0$$

$$11V_B - 7V_C - 4 = 0$$

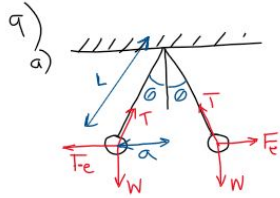
$$\frac{55}{3}V_C - 7V_C - 4 = 0$$

$$V_C = \frac{6}{17} = 0.353 \text{ V}$$

$$V_B = \frac{10}{17} = 0.588 \text{ V}$$

$$V_A = \frac{14}{17} = 0.824 \text{ V}$$

Homework



b) DISTANCE = $2a$ $L = 1 \text{ m}$
 $a = L \sin \theta$, $\theta = 5^\circ$
 $2a = 0.174 \text{ m}$

c) ① $T \cos \theta = \frac{m}{g}$
 ② $T \sin \theta = k \frac{Q^2}{(2a)^2}$

②/①: $\tan \theta = \frac{kQ^2}{(2a)^2} \frac{g}{m}$

$$Q = \sqrt{\frac{\tan \theta \cdot (2a)^2 m}{g k}}$$

$$= 7.75 \cdot 10^{-9} [\text{C}]$$

d) INCREASE

Homework

10) A vertical wire carries a current of 45 A due south. Calculate the magnitude and the direction of the magnetic field 2 cm to the right of the wire. **4.5×10^{-4} [T]**

11) A wire carries a current of 10 A. At what distance from the wire will a magnetic field of 8×10^{-4} [T] be produced? **2.5 mm**

Homework

12)

At $t = 0.27$ s, the magnet reaches the maximum induction voltage in the second coil. The speed of the magnet is then $v = g \cdot t = 9.81 \times 0.27 = 2.648$ m/s.

The induction voltage is then 1.35 V. The second coil contains 600 windings.

$$1.35 = \text{constant} \times 600 \times 2.648$$

$$\text{constant} = 8.49 \times 10^{-4} \text{ V-s-m}^{-1}.$$

At $t = 0.11$ s, the magnet reaches the maximum voltage in the first coil. The speed of the magnet is then $v = g \times t = 9.81 \times 0.11 = 1.079$ m/s.

The induction voltage is then 0.35 V.

$$0.35 = 8.49 \cdot 10^{-4} \times N \times 1.08$$

$$N = 3,817 \times 10^2$$

Rounded off, the first coil has 3.8×10^2 windings.

At $t = 0.36$ s, the magnet reaches maximum voltage in the third coil. The speed of the magnet is then $v = g \times t = 9.81 \times 0.36 = 3.531$ m/s

The induction voltage is then 0.80 V.

$$0.80 = 8.49 \cdot 10^{-4} \times N \times 3.53$$

$$N = 2,668 \times 10^2$$

Rounded off, the third coil has 2.7×10^2 windings.

The induction voltage is directly proportional to the flux change per unit time. So the induction voltage follows from the steepness of the tangent to the (flux, time) graph.

At $t = 0.11$ s, the steepness of the tangent to the graph is less steep than that at $t = 0.17$ s.

The induction voltage at $t = 0.11$ s is therefore smaller than that at $t = 0.17$ s.